

CASE-BASED SCENARIOS BY IDENTIFYING ENTITIES AND RELATIONSHIPS.

CSA0504

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1.CREATE AND PRESENT AN ER DIAGRAM FOR AN ONLINE LIBRARY SYSTEM, EXPLAINING EACH COMPONENT CLEARLY.

- Online Library System – ER Diagram
- Manages books and members
- Tracks borrowing and returning
- Stores library records
- Uses ER model for database design

ER DIAGRAM COMPONENTS

Entities: Real-world object (Book, Member)

- Member
- Book
- Librarian
- Loan

Attributes: Describes an entity

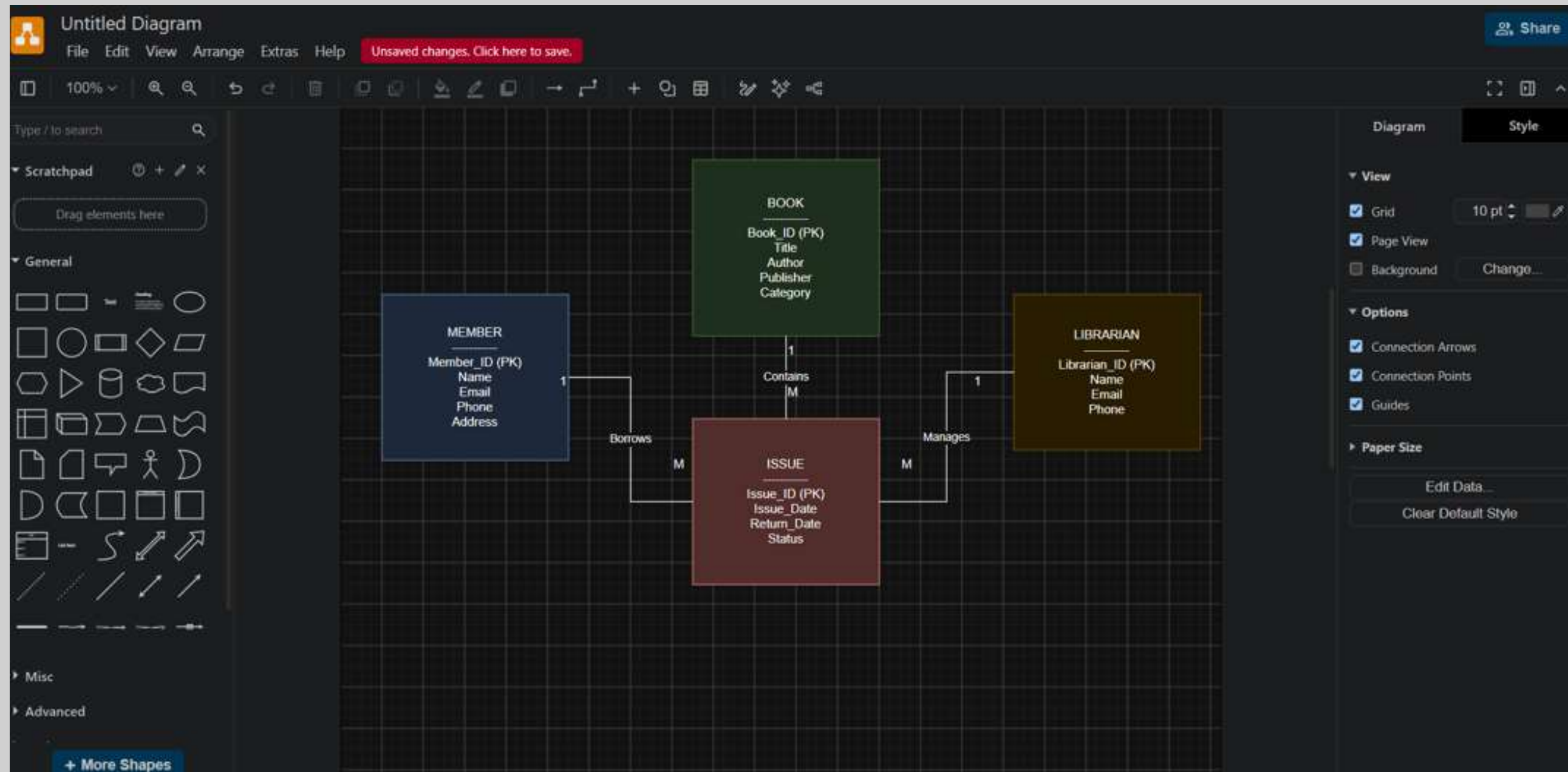
- Member ID, Name
- Book ID, Title
- Loan Date, Return Date

Relationships: Connects entities

- Member borrows Book
- Librarian manages Loan

Primary Key: Unique identifier (Book ID, Member ID)

ER DIAGRAM



2.PREPARE A PRESENTATION ON THE EER MODEL, DEMONSTRATING GENERALIZATION AND SPECIALIZATION WITH A UNIVERSITY CASE

- Enhanced ER (EER) Model
- Extension of ER model
- Supports inheritance
- Represents complex data
- Uses specialization and generalization

UNIVERSITY CASE STUDY

Superclass

- Person

Specialized Subclasses

- Student
- Faculty
- Staff

Each subclass inherits:

- Person ID
- Name
- Address

Additional attributes:

- Student → Semester
- Faculty → Department
- Staff → Designation

GENERALIZATION AND SPECIALIZATION

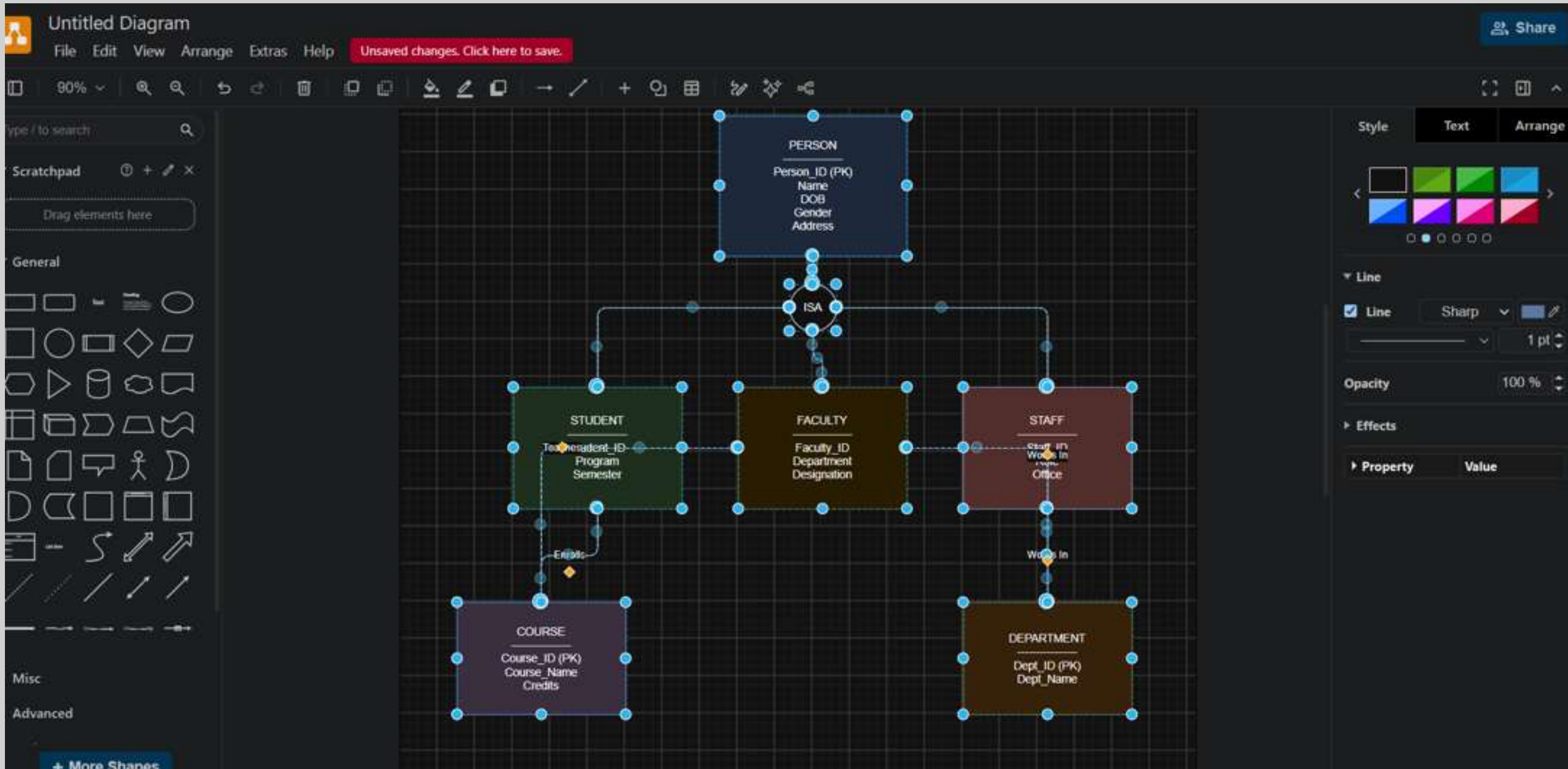
SPECIALIZATION

- Divides one entity into multiple subgroups.
- Example: Person → Student, Faculty, Staff.

GENERALIZATION

- Combines similar entities into one superclass.
- Reduces duplication of common attributes.
- Makes the database easier to maintain.
- Improves data organization.

EER DIAGRAM



3.PRESENT THE CONCEPT OF DATA INDEPENDENCE (LOGICAL AND PHYSICAL) AND ITS SIGNIFICANCE IN DBMS ARCHITECTURE.

Introduction

- Data independence means modifying one level of the database without affecting another.
- It is a key feature of DBMS architecture.
- It separates application programs from database changes.
- Supports easy database maintenance.
- Based on the three-schema architecture.
- Improves flexibility and scalability.

TYPES OF DATA INDEPENDENCE

Physical Data Independence

- Changes physical storage methods.
- Logical schema remains unchanged.
- Example: Changing file organization.

Logical Data Independence

- Changes logical database structure.
- User applications continue working.
- Example: Adding a new attribute.

IMPORTANCE OF DATA INDEPENDENCE

- Simplifies database maintenance.
- Reduces application modifications.
- Supports future database expansion.
- Improves system performance.
- Protects user applications from internal changes.
- Makes database management more efficient.



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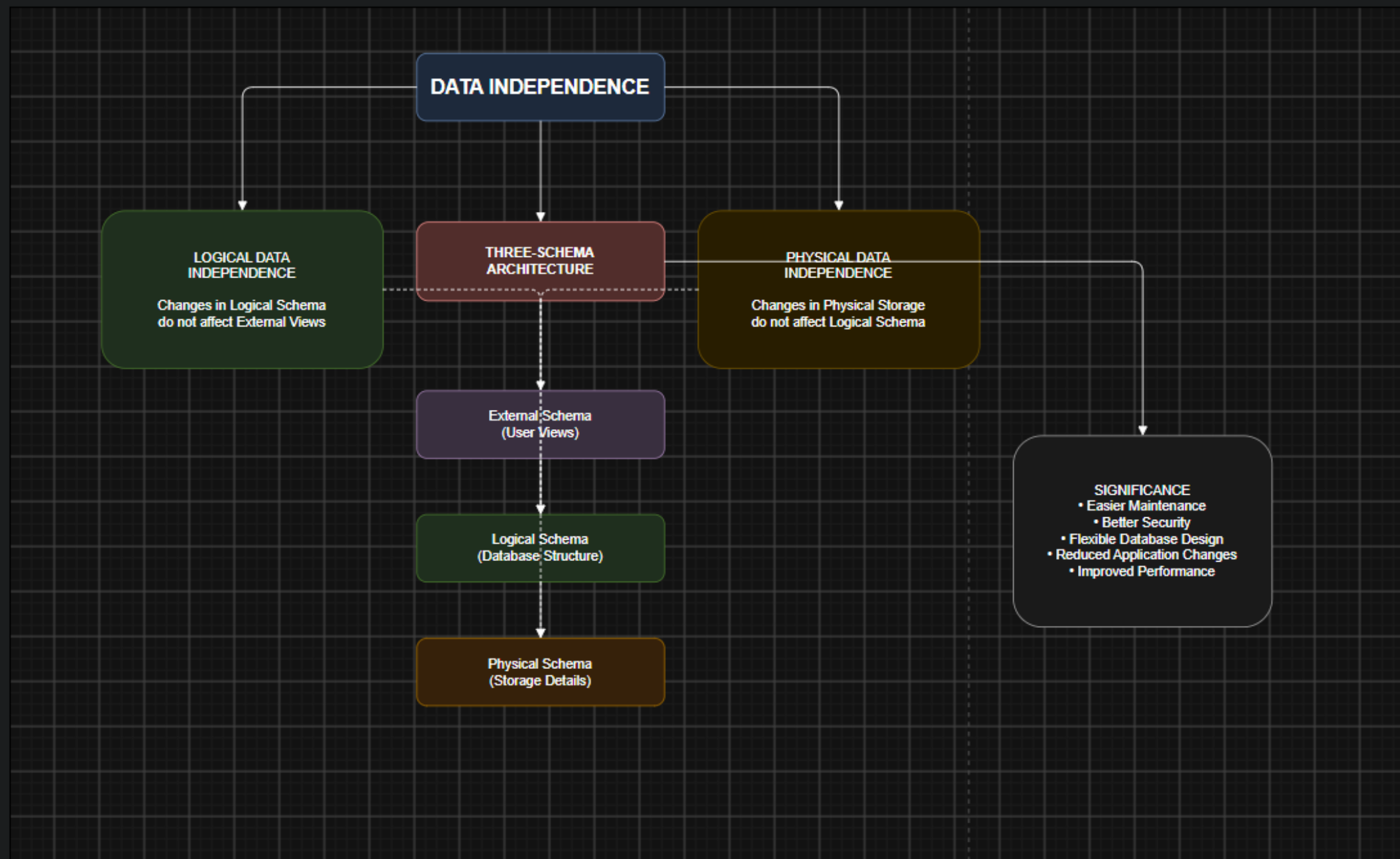
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4.DELIVER A TECHNICAL PRESENTATION ON THE SOFTWARE COMPONENTS OF A DBMS, EXPLAINING QUERY PROCESSOR, STORAGE MANAGER, AND TRANSACTION MANAGER.

- **DBMS consists of different software modules.**
- **These modules process, store, and manage data.**
- **The three major components are:**
- **Query Processor**
- **Storage Manager**
- **Transaction Manager**
- **Each component performs a specific function.**
- **Together they ensure efficient database operations.**

Query Processor

- Accepts SQL queries.
- Checks syntax and validity.
- Optimizes query execution.
- Retrieves requested data efficiently.

Storage Manager

- Manages files and indexes.
- Controls data storage on disks.
- Allocates memory space.
- Retrieves stored data quickly.

Transaction Manager

- Manages multiple database transactions.
- Ensures ACID properties.
- Maintains data consistency.
- Supports concurrent user access.
- Handles Commit and Rollback operations.
- Prevents data loss during failures.



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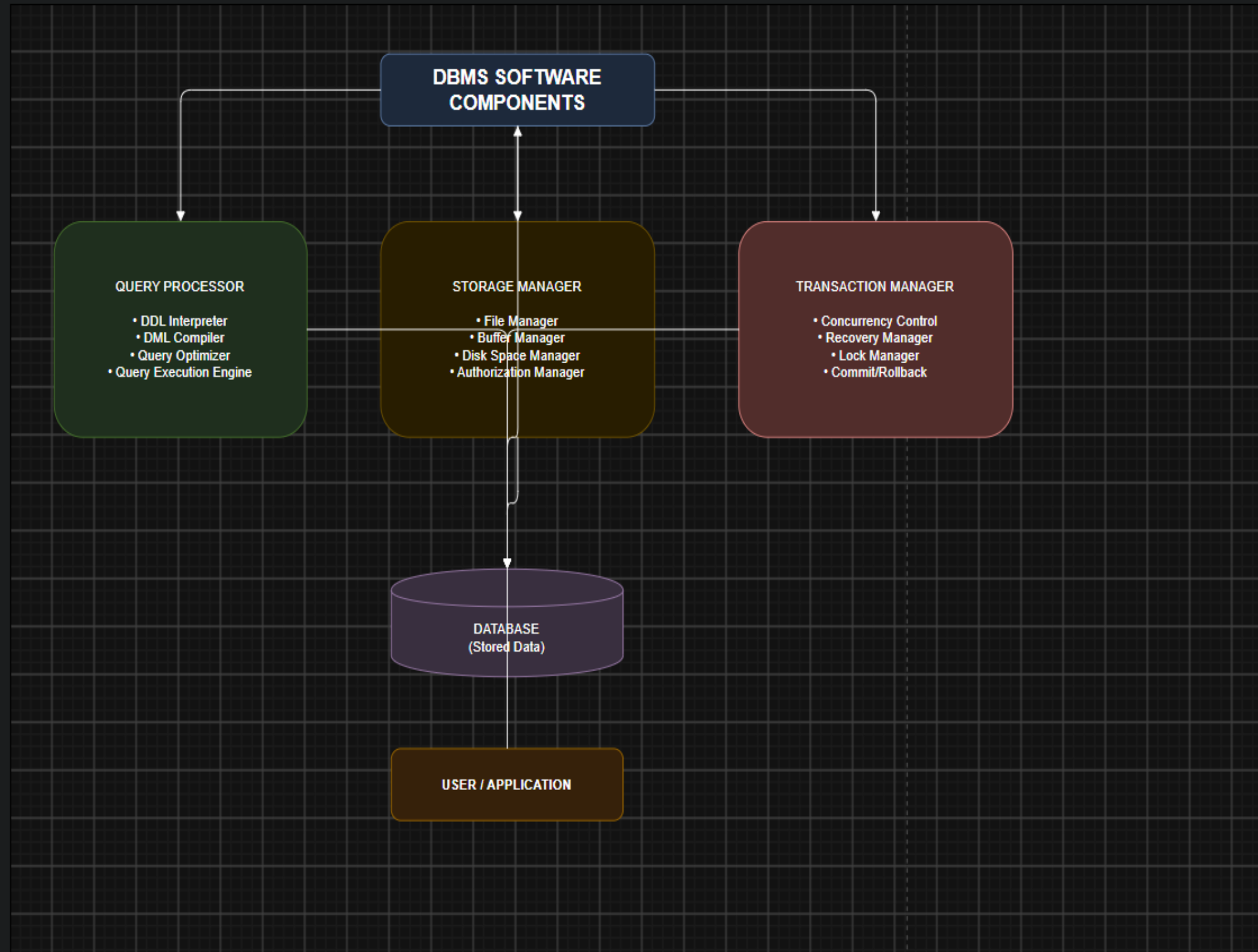
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5.PREPARE A PRESENTATION COMPARING STRONG AND WEAK ENTITIES IN ER MODELING WITH REAL-WORLD EXAMPLES.

Introduction

- Entities represent real-world objects in a database.
- They are classified as Strong or Weak entities.
- Strong entities exist independently.
- Weak entities depend on another entity.
- Both are represented differently in ER diagrams.
- They help define database relationships.

Strong Entity

Characteristics

- Has its own primary key.
- Exists independently.
- Represented by a single rectangle.
- Does not depend on another entity.

Examples

- Student
- Employee
- Book
- Customer

Weak Entity

Characteristics

- Has no complete primary key.
- Depends on a strong entity.
- Uses a partial key.
- Represented by a double rectangle.
- Connected using an identifying relationship.

Examples

- Employee → Dependent
- Order → Order Item
- Library Member → Loan Record



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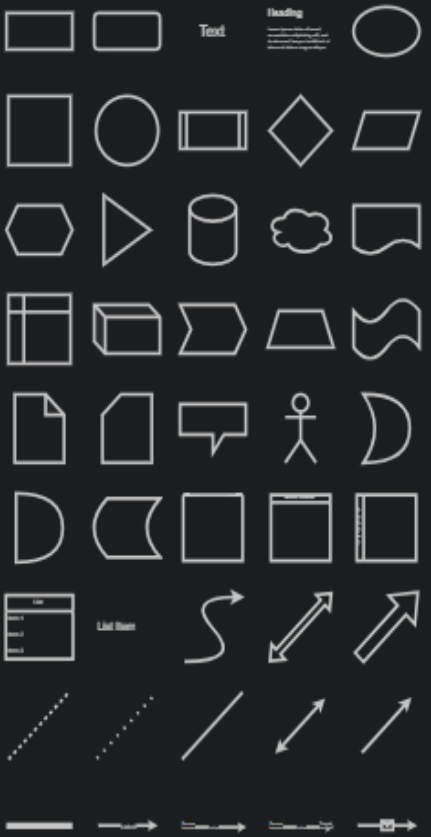


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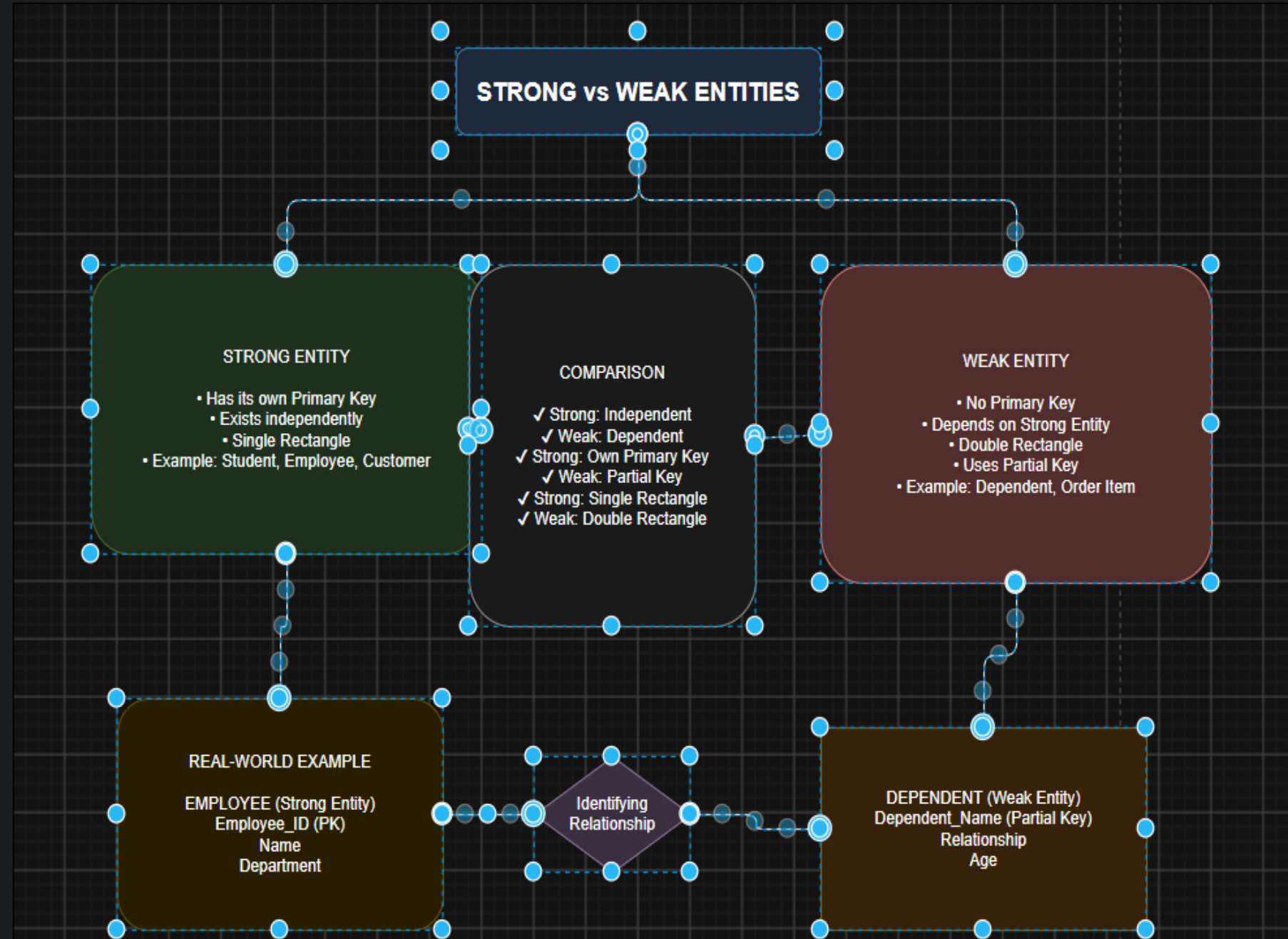
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